

Elimination of Branches (a) and (b) for the remaining degree pair (72, 108):

A Computational Audit of the GGHV Normal Form

Branch (a,b) Layer Reduction, Top-Layer Classification, and Final Obstruction

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Computation, Lean and LaTeX Source:

<https://drive.google.com/file/d/1NaoHgsXKao5cV-dDeQ760LoACN5WJGEH/view?usp=drivesdk>

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Abstract

We present a computational elimination of the branch (a,b) normal form arising from Proposition 4.3 of Guccione–Guccione–Horruitiner–Valqui [1] for a hypothetical (72, 108) counterexample to the planar Jacobian conjecture. Every base object is source-gated directly against GGHV v1 before any proof is drafted.

Starting from the verified Newton polygons

$$N(P) = \text{conv}\{(0, 0), (1, 0), (8, 14), (8, 16)\}, \quad N(Q) = \text{conv}\{(0, 0), (2, 1), (12, 21), (12, 24)\},$$

with $[P, Q] = x^2$ post- φ , we reduce the 72-unknown bilinear system to five one-variable layers $E_1, \dots, E_5$ totaling 92 equations, verified as an exact symbolic polynomial identity.

The top layer E_5 is the Davenport–Stothers extremal equation $\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta) = 1$ [2, 3]. The Belyi classification gives the upper bounds 3, 10, 35 chart points for $m = 3, 5, 7$; for $m = 7$ the bound is sharp, with the degree-35 eliminating polynomial $\mathcal{W}(a_6)$ irreducible over $\mathbb{Q}$, a quintic in $S = a_6^7$ with nonzero discriminant and exactly one real root. The 35 solutions are distinct and form a single Galois orbit: one real solution and 17 conjugate pairs, in five S -orbits. For $m = 7$ we also give an algebraic proof of this classification, valid over every field of characteristic 0. In reversed coordinates the chart equations are the coefficient recursion of $(1 + y_1v + \dots + y_7v^7)^{3/2}$. Six weighted-homogeneous relations, of weights 4, 5, 5, 6, 6, 7 and with explicit certificates, cut out the five torus orbits. We formalize this proof in Lean.

The descent proceeds through E_4 (an 18×19 matrix of rank 17, kernel dimension 2), E_3 (a 19×20 matrix of rank 18, kernel dimension 2, unconditionally solvable for every direction t), and E_2 (injective 19×12 , rank 12), leaving seven explicit compatibility conditions. We expand these as $b_i(t) + s_1M_i(t) + s_2L_i(t) = 0$ with b_i cubic and M_i, L_i linear, eliminate (s_1, s_2) to obtain 35 quintic 3×3 minors. The entire descent is computed exactly over the quintic field K_5 in characteristic 0: the 35 minors span all binary quintics, forcing $t = 0$. The case $t = 0$ forces B_0 constant, contradicting the required vertex (12, 24). The Lean formalization quantifies over all roots of R and all torus parameters, covering all 35 top-layer solutions.

Hence no polynomials P, Q satisfying NewtonNF2 have $[P, Q] = \lambda x^2$ with $\lambda \neq 0$ (Theorem 1.1). In Lean this NewtonNF2 form is a single unconditional kernel-checked theorem, `main_theorem`, on the standard axioms only. Its top-layer input is `chartClassification_holds`, a formalized algebraic proof of the $m = 7$ top-layer classification

of Proposition 6.1, in the normalized chart (§6.4). The $a_{8,16} = 0$ part is computer algebra. The application to the Jacobian conjecture is conditional on the correctness and exhaustiveness of [1, Proposition 4.3] (Corollary 1.2).

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1 Introduction

1.1 Context and prior work

The planar Jacobian conjecture asserts that a polynomial map $(P, Q) : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ with $[P, Q] := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ is invertible. In [1], Guccione, Guccione, Horruitiner, and Valqui raise the lower bound on $\max\{\deg P, \deg Q\}$ for a hypothetical counterexample from 100 (Moh [4]) to 108, leaving open only the degree pair $(72, 108)$ and its symmetric $(108, 72)$.

Santibáñez-Leal has published machine evidence related to the $(72, 108)$ candidate [5, 6], explicitly not claiming elimination. The candidate itself is documented in [7].

For the degree pair $(72, 108)$, [1] (an unrefered preprint, arXiv:2204.14178v1) identifies two cases. The corner $(9, 27)$ is discarded in [1, §5]; the open case is the corner $(8, 28)$ with $(m, n) = (3, 2)$. For this case, [1, Proposition 4.3] shows that any counterexample gives rise, after an explicit (non-invertible) morphism φ , to one of two Newton polygon configurations. Case (1), with additional vertices $(0, 8)$ resp. $(0, 12)$, is branch (c). Case (2), the subject of this paper, is branches (a) and (b):

$$N(P) = \text{conv}\{(0, 0), (1, 0), (8, 14), (8, 16)\}, \quad N(Q) = \text{conv}\{(0, 0), (2, 1), (12, 21), (12, 24)\}, \quad (1)$$

with $[P, Q] = x^2$ in post- φ coordinates. See Figure 1 for the polygon shapes.

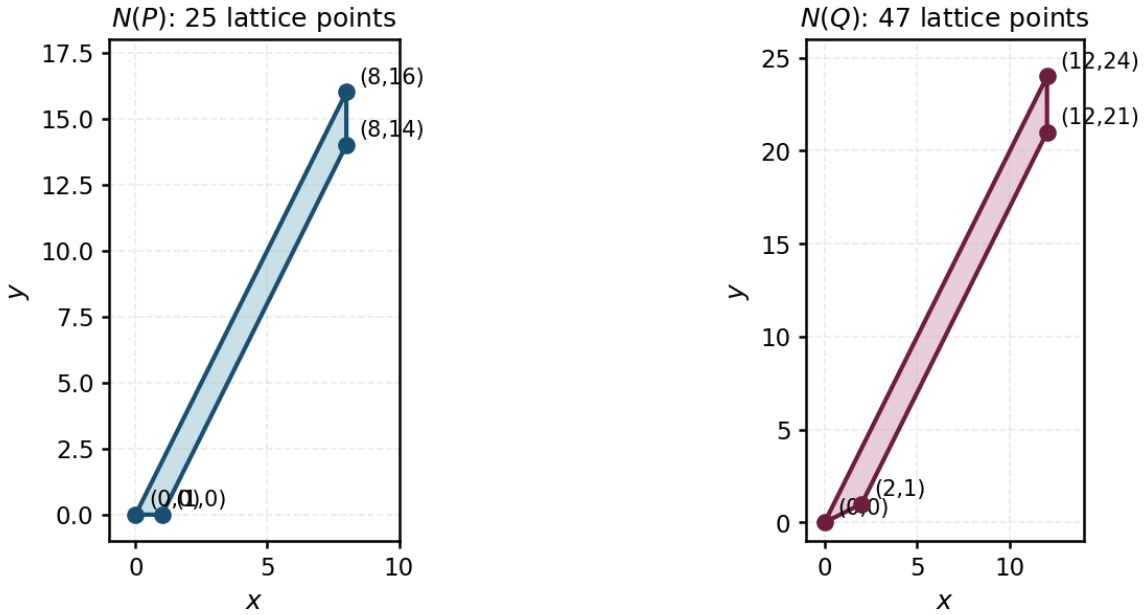


Figure 1: The branch (a,b) Newton polygons from [1, Proposition 4.3(2)], source-gated at v1 lines 495 and 599–600. $N(P)$ has 25 lattice points; $N(Q)$ has 47.

1.2 Main result

Theorem 1.1 (Elimination of branches (a) and (b)). *Let K be a field of characteristic 0. Let $P, Q \in K[x, y]$ have $\text{supp } P \subset N_P$ and $\text{supp } Q \subset N_Q$, with $a_{1,0} \cdot a_{8,14} \cdot b_{2,1} \cdot b_{12,21} \neq 0$ and $[P, Q] = \lambda x^2$ for some $\lambda \neq 0$. Then $b_{12,24} = 0$ and $a_{8,16} = 0$. In particular, no P, Q satisfy $\text{NewtonNF2}(P, Q)$ (listing 11; the polygons are in §2) with $[P, Q] = \lambda x^2$.*

Corollary 1.2 (Conditional on GGHV). *Assuming [1, Prop. 4.3], no counterexample to the plane Jacobian conjecture arises in case $(8, 28)$, sub-cases a) and b).*

The proof is computational but exact. We reduce the 72-unknown system to triangular layers (Figure 2), determine the top layer via the Belyi-map/Riemann-existence argument (Figure 3), and independently by an algebraic argument formalized in Lean (§6.4). We then obstruct the descent via the Lean-formalized layer certificates (§7). Key steps were verified in SINGULAR 4.4.1 and Python/SYMPY; scripts are archived with this draft.

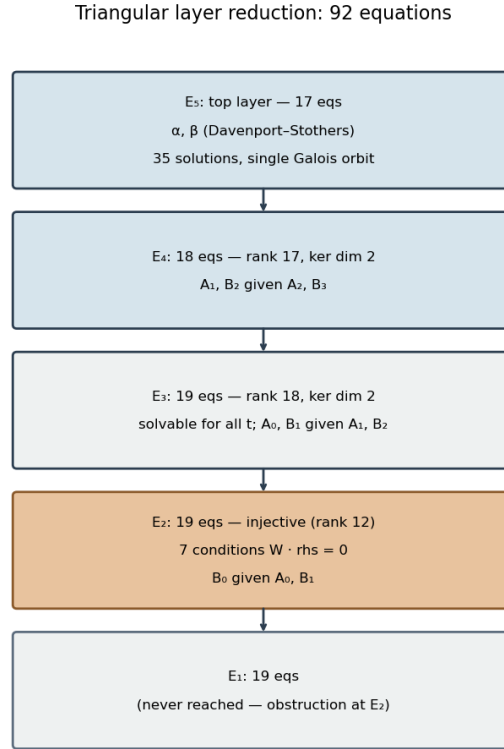


Figure 2: The triangular layer structure: 92 equations in five layers. Each layer is affine-linear in its unknowns given the solution above it.

1.3 Logical status

Remark 1.3. Theorem 1.1 does not depend on GGHV. Its proof uses Proposition 6.1 for $m = 7$, which is proved twice. On paper it follows from Riemann’s existence theorem (§6). In §6.4 the classification it provides is proved algebraically, in the normalized chart, and that proof is formalized in Lean as `chartClassification_holds` (listing 16). In Lean, `main_theorem_of_chart` (listing 7) derives the `NewtonNF2` form of Theorem 1.1 from the explicit statement `ChartClassification` (listing 12). `main_theorem` (listing 17) combines it with `chartClassification_holds` into a single unconditional kernel-checked theorem, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The $b_{12,24} = 0$ part follows from `no_completion_K5_PQ` together with `topLayerClassification_of_chart` and `chartClassification_holds`; the $a_{8,16} = 0$ part is computer algebra (the exact Gröbner computation `a816_full.sing`, not Lean; see §5). Only Corollary 1.2 also assumes GGHV Proposition 4.3.

1.4 Outline

Section 2 source-gates all base objects against GGHV v1. Section 3 sets up the 72-unknown system. Section 4 gives the exact layer reduction. Section 5 fixes normalizations. Section 6 classifies the top layer. Section 7 carries out the descent. Section 8 proves the final obstruction. Section 9 lifts to characteristic zero. Section 10 concludes and discusses the remaining branch (c). Section 11 covers prior work, attribution, and independence.

2 Source data

We verify every base object directly against version 1 of [1] (arXiv:2204.14178v1), downloaded from <https://arxiv.org/pdf/2204.14178v1> and extracted with `pdftotext -layout`. No proof is drafted until the polygons, bracket, and φ are confirmed character-by-character against GGHV's text.

Proposition 2.1 ([1, Proposition 4.3], source-gated). *If there is a counterexample to the Jacobian conjecture in case $(8, 28)$, then there exist $P, Q \in L^{(1)}$ with $[P, Q] = x^2$ and one of:*

- (1) $N(P) = \{(0, 0), (1, 0), (8, 14), (8, 16), (0, 8)\}$,
 $N(Q) = \{(0, 0), (2, 1), (12, 21), (12, 24), (0, 12)\}$;
- (2) $N(P) = \{(0, 0), (1, 0), (8, 14), (8, 16)\}$,
 $N(Q) = \{(0, 0), (2, 1), (12, 21), (12, 24)\}$.

Source locations. The `pdftotext -layout` extraction of v1 yields:

- Line 33: $[P, Q] := P_x Q_y - P_y Q_x \in K^\times$ (bracket convention).
- Line 492: "Proposition 4.3 (Case $(8, 28)$)."
- Line 493: "with $[P, Q] = x^2$ " (rendered as x^2 in the extraction).
- Line 495: Case (2) polygons
 $N(P) = \{(0, 0), (1, 0), (8, 14), (8, 16)\}$, $N(Q) = \{(0, 0), (2, 1), (12, 21), (12, 24)\}$.
- Lines 596–597: The morphism $\varphi(x) = x^{-1}$, $\varphi(y) = x^4 y$; "this is not an automorphism and the chain rule gives $[\varphi(P), \varphi(Q)] = -[P, Q]x^2$."
- Lines 599–600: "in the cases a) and b) the Newton Polygons of P and Q become $N(P) = \{(0, 0), (1, 0), (8, 14), (8, 16)\}$, $N(Q) = \{(0, 0), (2, 1), (12, 21), (12, 24)\}$."

A verification script (`source_gate_verify.py`) asserts each of these lines automatically; it exits non-zero if any check fails. $\square$

We work exclusively with case (2) (branches a,b).

3 The polynomial system

3.1 Lattice points

The Newton polygons (1) contain:

$$|N(P) \cap \mathbb{Z}^2| = 25, \quad |N(Q) \cap \mathbb{Z}^2| = 47, \quad (2)$$

verified by direct enumeration (cross-checked with Pick's theorem: $\text{area}(N(P)) = 15$, boundary 18, interior 7). This gives $25 + 47 = 72$ unknown coefficients. The two constant

terms do not appear in $[P, Q]$, leaving 70 effective unknowns, but we retain all 72 for the layer structure.

3.2 The bracket equations

Writing $P = \sum_{(i,j) \in N(P)} a_{ij} x^i y^j$ and $Q = \sum_{(i,j) \in N(Q)} b_{ij} x^i y^j$, the condition $[P, Q] = x^2$ expands to bilinear equations in the coefficients. Solving the full 72-unknown bilinear system directly is impractical; we use the exact layer reduction of §4. (Direct expansion remains feasible for *verification* of the reduction, as in Proposition 4.1.)

4 Exact layer reduction

Set $u = xy^2$ and decompose into y -layers:

$$P = \sum_k y^{-k} A_k(u), \quad Q = \sum_\ell y^{-\ell} B_\ell(u), \quad (3)$$

where $A_k, B_\ell \in \mathbb{C}[u]$ are one-variable polynomials whose coefficients are linear forms in the a_{ij}, b_{ij} . The bracket $[P, Q]$ becomes a sum over y -layers, each a one-variable identity in u .

Proposition 4.1 (Exact layer reduction). *The system $[P, Q] = x^2$ on the 72 coefficients implies five one-variable polynomial identities $E_1, \dots, E_5$ (the converse is not needed):*

Layer	Equations	Unknowns (given previous layers)
E_5	17	Top: α, β
E_4	18	A_1, B_2 given A_2, B_3
E_3	19	A_0, B_1 given A_1, B_2
E_2	19	B_0 given A_0, B_1
E_1	19	Lowest coefficients
Total	92	

The layer reduction was verified as a polynomial identity (script `C1_layer_reduction.sing`) with all 72 coefficients treated as symbolic indeterminates; the residual is identically zero.

Computational verification. Substituting (3) into $[P, Q] - x^2$ and collecting by powers of y yields the five identities $E_1, \dots, E_5$. Expanding each E_i in the 72 indeterminates and comparing coefficients with the direct bilinear expansion of $[P, Q] - x^2$ gives zero difference as a polynomial in 72 variables. Script: `C1_layer_reduction.sing` (SINGULAR). $\square$

Listing 1: Mechanically extracted from `Jacobian/BranchAbLayers.lean` (sha256 7c16141b4b126b3e); statement verbatim, proof elided.

```
theorem layers_of_jac (lam : K) (A0 A1 A2 B0 B1 B2 B3 : Polynomial K)
  (p0 p1 p2 q0 q1 q2 q3 : MvPolynomial (Fin 2) K)
  (hp0 : X 1 ^ 0 * p0 = evH A0) (hp1 : X 1 ^ 1 * p1 = evH A1) (hp2 : X 1 ^ 2 * p2 =
    ↪ evH A2)
  (hq0 : X 1 ^ 0 * q0 = evH B0) (hq1 : X 1 ^ 1 * q1 = evH B1) (hq2 : X 1 ^ 2 * q2 =
    ↪ evH B2)
  (hq3 : X 1 ^ 3 * q3 = evH B3)
```

```

(hJ : jac (p0 + p1 + p2) (q0 + q1 + q2 + q3) = C lam * X 0 ^ 2) :
layerTerm 2 3 A2 B3 = Polynomial.C lam * Polynomial.X ^ 2 ^
layerTerm 2 2 A2 B2 + layerTerm 1 3 A1 B3 = 0 ^
layerTerm 2 1 A2 B1 + layerTerm 1 2 A1 B2 + layerTerm 0 3 A0 B3 = 0 ^
layerTerm 2 0 A2 B0 + layerTerm 1 1 A1 B1 + layerTerm 0 2 A0 B2 = 0 ^
layerTerm 1 0 A1 B0 + layerTerm 0 1 A0 B1 = 0 := by
-- proof body elided (see artifact)

```

The basic definitions used above are printed verbatim here, since only a human reader (not the Lean kernel) checks definitions:

Listing 2: Mechanically extracted from Jacobian/BranchAbLayers.lean (sha256 7c16141b4b126b3e); definition printed verbatim in full.

```

def jac (P Q : MvPolynomial (Fin 2) K) : MvPolynomial (Fin 2) K :=
  pderiv 0 P * pderiv 1 Q - pderiv 1 P * pderiv 0 Q

```

Listing 3: Mechanically extracted from Jacobian/BranchAbLayers.lean (sha256 7c16141b4b126b3e); definition printed verbatim in full.

```

def uu : MvPolynomial (Fin 2) K := X 0 * X 1 ^ 2

```

Listing 4: Mechanically extracted from Jacobian/BranchAbLayers.lean (sha256 7c16141b4b126b3e); definition printed verbatim in full.

```

def evH : Polynomial K →a[K] MvPolynomial (Fin 2) K := Polynomial.aeval (uu (K := K))

```

Listing 5: Mechanically extracted from Jacobian/BranchAbLayers.lean (sha256 7c16141b4b126b3e); definition printed verbatim in full.

```

def layerTerm (a b : ℕ) (A B : Polynomial K) : Polynomial K :=
  (a : Polynomial K) * A * Polynomial.derivative B - (b : Polynomial K) *
  ↪ Polynomial.derivative A * B

```

The system is triangular (Figure 2): E_5 involves only the top-layer unknowns, and each E_i ($i < 5$) is affine-linear in its unknowns given a solution of the layers above. This is the key structural fact that makes the problem tractable.

5 Normalization

The top-layer unknowns admit rescaling symmetries from the torus action. We fix, without loss of generality,

$$a_{1,0} = a_{8,14} = 1, \quad (4)$$

leaving $a_{8,16}$ free. The remaining 1-dimensional torus acts by the Y -weight grading used in the layer decomposition (§4); the finite subgroup μ_7 gives 7 chart copies per orbit.

Remark 5.1 (Lean chart normalization). The Lean formalization uses chart coordinates $(a_1, \dots, a_7, b_0, \dots, b_9)$ with the torus normalization $a_7^3 b_0^2 = 1$ (see `ChartClassification` in `Jacobian/BranchAbChart.lean`). After fixing $\alpha_0 = 1$ and $\beta_{10} = 1$, the remaining scaling $u \mapsto \kappa u$ acts on a_7 with weight 7 and on b_0 with weight -10 , hence on $a_7^3 b_0^2$ with weight $21 - 20 = 1$. The condition $a_7^3 b_0^2 = 1$ therefore fixes κ completely, with no 7th-root ambiguity: the Lean chart meets each torus orbit *exactly once*, giving 5 points, not 35. The §5 chart ($a_{1,0} = a_{8,14} = 1$) meets each orbit 7 times. The two charts are related by a torus element requiring a 7th root.

Vertex conditions used. The proof uses $(1, 0)$ and $(2, 1)$ (equivalently $\lambda \neq 0$), $(8, 14)$ (normalization and top-layer degree), and $(12, 24)$ (the $t = 0$ exclusion). The vertex $(8, 16)$ is left free in the Lean formalization; however, an independent Gröbner computation (script `a816_full.sing`, rerunnable from the computational archive) shows that $a_{8,16} = 0$ is forced once E_1 is included, giving a second independent contradiction from the lower edge.

Listing 6: Mechanically extracted from `Jacobian/BranchAbTorus.lean` (sha256 97918673d5c4dc09); statement verbatim, proof elided.

```
theorem layers_transport (α β κ : K) (A₀ A₁ A₂ B₀ B₁ B₂ B₃ : K[X])
  (E4 : layerTerm 2 2 A₂ B₂ + layerTerm 1 3 A₁ B₃ = 0)
  (E3 : layerTerm 2 1 A₂ B₁ + layerTerm 1 2 A₁ B₂ + layerTerm 0 3 A₀ B₃ = 0)
  (E2 : layerTerm 2 0 A₂ B₀ + layerTerm 1 1 A₁ B₁ + layerTerm 0 2 A₀ B₂ = 0) :
  layerTerm 2 2 (C α * sc κ A₂) (C β * sc κ B₂) + layerTerm 1 3 (C α * sc κ A₁) (C
    ↪ β * sc κ B₃) = 0 ∧
  layerTerm 2 1 (C α * sc κ A₂) (C β * sc κ B₁) + layerTerm 1 2 (C α * sc κ A₁) (C
    ↪ β * sc κ B₂) +
    layerTerm 0 3 (C α * sc κ A₀) (C β * sc κ B₃) = 0 ∧
  layerTerm 2 0 (C α * sc κ A₂) (C β * sc κ B₀) + layerTerm 1 1 (C α * sc κ A₁) (C
    ↪ β * sc κ B₁) +
    layerTerm 0 2 (C α * sc κ A₀) (C β * sc κ B₂) = 0 := by
  -- proof body elided (see artifact)
```

6 The top layer

6.1 The Davenport–Stothers equation

The top identity E_5 takes the form

$$\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta) = 1, \quad (5)$$

where $\alpha = A_2/u$ and $\beta = B_3/u^2$ are polynomials of degree 7 and 10 respectively, the degrees being forced by the vertices $(8, 14) \in N(P)$ and $(12, 21) \in N(Q)$.

Setting $u = s^2$ with $f(s) = \alpha(s^2)$ and $g(s) = s\beta(s^2)$, equation (5) becomes the Davenport–Stothers extremal equation [2, 3]

$$2fg' - 3f'g = c \neq 0. \quad (6)$$

Indeed $2fg' - 3f'g = 2(\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta))$. With $\deg f = 14 = 2k$ and $\deg g = 21 = 3k$ ($k = 7$), we have $\deg(g^2 - c \cdot f^3) = 8 = k + 1$, which is exactly Davenport’s bound: the pair is extremal. with f even and g odd. Here $\deg_s f = 2m$ where $m = \deg_u \alpha$; the relation $1 + 2n - 3m = 0$ (with $n = \deg_u \beta$) forces m odd. The cases $m = 1, 3, 5, 7$ admit solutions; $m = 7$ is the relevant case for (1). The symmetric (parity-constrained) classification is computed here via elimination; we do not claim it follows from the literature.

6.2 Classification

Proposition 6.1 (Top-layer solutions). *The system E_5 has:*

- $m = 3$: 3 complex solutions; eliminant $3a_2^3 - 32$, irreducible over $\mathbb{Q}$;
- $m = 5$: 10 complex solutions; eliminant $9a_4^{10} + 37200a_4^5 + 95051008$, irreducible over $\mathbb{Q}$;
- $m = 7$: 35 complex solutions; eliminant $\mathcal{W}(a_6)$ of degree 35.

The $m = 7$ case is the one relevant to (1).

Proof. We work over $\mathbb{C}$: the Belyi classification is topological, hence valid over $\mathbb{C}$, and the resulting count of solutions transfers to any field of characteristic 0 by base change (the equations are defined over $\mathbb{Q}$). We normalize the third branch point to $c = 1$ by the scaling $u \mapsto cu$, which preserves the torus action. We prove the counts in characteristic 0 via the Belyi-map classification. For the top layer, E_5 is $\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta) = 1$ with $\deg \alpha = m$, $\deg \beta = (3m-1)/2$ ($m = 3, 5, 7$, so $\deg \beta = 4, 7, 10$). Put $\varphi = u\beta^2/\alpha^3$. Then $\varphi' = \beta E/\alpha^4 = \beta/\alpha^4$, so φ is a Belyi map: it ramifies only over 0, ∞ , and $c = \varphi(\infty) \neq 0$. The ramification is forced as follows. At a root r of α , $E(r) = -3r \cdot \alpha'(r)\beta(r) = 1$. So α has simple nonzero roots, none shared with β . At a root s of β , $E(s) = 2s \cdot \alpha(s)\beta'(s) = 1$, so β also has simple nonzero roots. Hence $\deg \varphi = 3m$, and $\varphi' = \beta/\alpha^4$ vanishes in $\mathbb{C}$ only at the roots of β , which forces the passport

$$\begin{array}{l|ll} m = 3 & [2^4 1] & (n = 9) & [3^3] & [7 \cdot 1^2] \\ m = 5 & [2^7 1] & (n = 15) & [3^5] & [12 \cdot 1^3] \\ m = 7 & [2^{10} 1] & (n = 21) & [3^7] & [17 \cdot 1^4] \end{array}$$

By Riemann's existence theorem, covers with this passport correspond to S_n -conjugacy classes of transitive triples in $C_0 \times C_1 \times C_\infty$ with product 1 (here $n = 3m$). The Frobenius character formula (computed exactly in `belyi_counts_m357.py` via the Murnaghan–Nakayama rule) counts N triples with $N = 9!, 2 \cdot 15!, 5 \cdot 21!$ for $m = 3, 5, 7$; since every such triple has trivial centralizer, each S_n -conjugacy class has $n!$ elements, giving $N/n! = 1, 2, 5$ covers. Every triple counted is transitive, for general m : a non-transitive triple would have an orbit on which g_∞ acts trivially, so on that orbit $g_0 = g_1^{-1}$; but g_1 has cycle type 3^m (hence fixes no point, and so does g_1^{-1}), while g_0 has cycle type $2^{(3m-1)/2} 1$ and contains no 3-cycle—a contradiction. Hence all counted triples are genuine covers. Each cover has trivial automorphism group: an automorphism $u \mapsto \kappa u$ acts freely on the m poles and on the $(m+1)/2$ unramified points over c , so its order divides $\gcd(m, (m+1)/2) = 1$.

Now suppose two solutions give isomorphic covers. The isomorphism fixes the unique unramified point over 0 ($u = 0$) and the unique maximally ramified point over c ($u = \infty$). So it is $u \mapsto \kappa u$, and comparing zeros and poles shows the two solutions differ by the torus action. Hence distinct orbits give non-isomorphic covers, and there are at most $m \cdot (\# \text{covers})$ chart points: 3, 10, 35.

For $m = 7$, the exactly verified parametrization (`script verify_msolve_param.py`, shipped with inputs `e5_m7.ms` and `e5_m7.out` in `scripts/`) exhibits 35 distinct points, so the classification is complete in characteristic 0. For $m = 3$ and $m = 5$ the Belyi bound is likewise sharp. In the normalization $\alpha_0 = \beta_0 = \alpha_m = 1$ the system is zero-dimensional, of vector-space dimension 3 for $m = 3$ and 10 for $m = 5$; eliminating all variables but a_{m-1} gives, up to a nonzero rational factor, the stated eliminant. Each eliminant is irreducible and separable over $\mathbb{Q}$, so its d distinct roots are all realized as a_{m-1} -coordinates of solutions (the system is defined over $\mathbb{Q}$, hence Galois-stable), giving at least d distinct solutions; with the Belyi upper bound of d , there are exactly d . All of this is checked by `script e5_exact_counts_m35.py` in `scripts/`.

The $m = 3$ eliminant $3a_2^3 - 32$ and $m = 5$ eliminant $9a_4^{10} + 37200a_4^5 + 95051008$ are verified

irreducible over $\mathbb{Q}$ by factorization (same script). For $m = 7$, the eliminating polynomial in a_6 has degree 35 (Gröbner elimination via SINGULAR eliminate). $\square$

Remark 6.2 (Status of ChartClassification). ChartClassification (in Jacobian/BranchAbChart.lean) encodes the 17 equations of the E_5 chart system: the 16 bilinear equations plus the torus normalization $a_7^3 b_0^2 = 1$. Up to v16 of the artifact it was an unproven hypothesis standing for the case $m = 7$ of Proposition 6.1. It is now a theorem, chartClassification_holds (§6.4), so the implication ChartClassification $L \rightarrow \neg \exists \dots$ of main_theorem_of_chart is discharged and cannot be vacuous. The statement itself is unchanged. Its 17 equations were verified against the chart_system_Q.sing Singular script and by the mechanical chart audit (see the computational archive). The generator of the proof in §6.4 reads this same statement from the Lean source.

Listing 7: Mechanically extracted from Jacobian/BranchAbChart.lean (sha256 15907ffeffb6d274); statement verbatim, proof elided.

```
theorem main_theorem_of_chart {L : Type*} [Field L] [CharZero L] (hc :
  ⇔ ChartClassification L) :
  ¬ ∃ (P Q : MvPolynomial (Fin 2) L) (lam : L), lam ≠ 0 ∧ NewtonNF2 P Q ∧ jac P Q =
    ⇔ MvPolynomial.C lam * MvPolynomial.X 0 ^ 2 :=
  -- proof term elided (see artifact)
```

The two definitions on which main_theorem_of_chart rests are printed verbatim in full:

Listing 8: Mechanically extracted from Jacobian/BranchAbNewton.lean (sha256 fda6fce4c16abbe8); definition printed verbatim in full.

```
abbrev mono (i j : ℕ) : Fin 2 →0 ℕ := Finsupp.single 0 i + Finsupp.single 1 j

@[simp] lemma mono_apply_zero (i j : ℕ) : mono i j 0 = i := by simp [mono]
@[simp] lemma mono_apply_one (i j : ℕ) : mono i j 1 = j := by simp [mono]
```

Listing 9: Mechanically extracted from Jacobian/BranchAbNewton.lean (sha256 fda6fce4c16abbe8); definition printed verbatim in full.

```
def inNP (m : Fin 2 →0 ℕ) : Prop := m 0 ≤ 8 ∧ m 1 ≤ 2 * m 0 ∧ 2 * m 0 ≤ m 1 + 2
```

Listing 10: Mechanically extracted from Jacobian/BranchAbNewton.lean (sha256 fda6fce4c16abbe8); definition printed verbatim in full.

```
def inNQ (m : Fin 2 →0 ℕ) : Prop := m 0 ≤ 12 ∧ m 1 ≤ 2 * m 0 ∧ 2 * m 0 ≤ m 1 + 3 ∧ m
  ⇔ 0 ≤ 2 * m 1
```

Listing 11: Mechanically extracted from Jacobian/BranchAbNewton.lean (sha256 fda6fce4c16abbe8); definition printed verbatim in full.

```

def NewtonNF2 (P Q : MvPolynomial (Fin 2) K) : Prop :=
  (∀ m ∈ P.support, inNP m) ∧ (∀ m ∈ Q.support, inNQ m) ∧
  P.coeff (mono 0 0) ≠ 0 ∧ P.coeff (mono 1 0) ≠ 0 ∧ P.coeff (mono 8 14) ≠ 0 ∧ P.coeff
  ↪ (mono 8 16) ≠ 0 ∧
  Q.coeff (mono 0 0) ≠ 0 ∧ Q.coeff (mono 2 1) ≠ 0 ∧ Q.coeff (mono 12 21) ≠ 0 ∧
  ↪ Q.coeff (mono 12 24) ≠ 0

```

Listing 12: Mechanically extracted from Jacobian/BranchAbChart.lean (sha256 15907ffeffb6d274); definition printed verbatim in full.

```

def ChartClassification (L : Type*) [Field L] : Prop :=
  ∀ a1 a2 a3 a4 a5 a6 a7 b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 : L,
  (3 : L) * b1 + (-2 : L) * a1 * b0 = 0 →
  (5 : L) * b2 + (-5 : L) * a2 * b0 = 0 →
  (7 : L) * b3 + (2 : L) * a1 * b2 + (-3 : L) * a2 * b1 + (-8 : L) * a3 * b0 = 0 →
  (9 : L) * b4 + (4 : L) * a1 * b3 + (-1 : L) * a2 * b2 + (-6 : L) * a3 * b1 + (-11
  ↪ : L) * a4 * b0 = 0 →
  (11 : L) * b5 + (6 : L) * a1 * b4 + (1 : L) * a2 * b3 + (-4 : L) * a3 * b2 + (-9
  ↪ : L) * a4 * b1 + (-14 : L) * a5 * b0 = 0 →
  (13 : L) * b6 + (8 : L) * a1 * b5 + (3 : L) * a2 * b4 + (-2 : L) * a3 * b3 + (-7
  ↪ : L) * a4 * b2 + (-12 : L) * a5 * b1 + (-17 : L) * a6 * b0 = 0 →
  (15 : L) * b7 + (10 : L) * a1 * b6 + (5 : L) * a2 * b5 + (-5 : L) * a4 * b3 +
  ↪ (-10 : L) * a5 * b2 + (-15 : L) * a6 * b1 + (-20 : L) * a7 * b0 = 0 →
  (17 : L) * b8 + (12 : L) * a1 * b7 + (7 : L) * a2 * b6 + (2 : L) * a3 * b5 + (-3
  ↪ : L) * a4 * b4 + (-8 : L) * a5 * b3 + (-13 : L) * a6 * b2 + (-18 : L) * a7 *
  ↪ b1 = 0 →
  (19 : L) * b9 + (14 : L) * a1 * b8 + (9 : L) * a2 * b7 + (4 : L) * a3 * b6 + (-1
  ↪ : L) * a4 * b5 + (-6 : L) * a5 * b4 + (-11 : L) * a6 * b3 + (-16 : L) * a7 *
  ↪ b2 = 0 →
  (21 : L) * a1 * b9 + (16 : L) * a2 * b8 + (11 : L) * a3 * b7 + (6 : L) * a4 * b6 + (-4 : L) * a5 * b5 + (-9 : L) * a6 * b4 + (-14 : L) * a7 * b3 = 0 →
  (18 : L) * a1 + (13 : L) * a2 * b9 + (8 : L) * a3 * b8 + (3 : L) * a4 * b7 + (-2
  ↪ : L) * a5 * b6 + (-7 : L) * a6 * b5 + (-12 : L) * a7 * b4 = 0 →
  (15 : L) * a2 + (10 : L) * a3 * b9 + (5 : L) * a4 * b8 + (-5 : L) * a6 * b6 +
  ↪ (-10 : L) * a7 * b5 = 0 →
  (12 : L) * a3 + (7 : L) * a4 * b9 + (2 : L) * a5 * b8 + (-3 : L) * a6 * b7 + (-8
  ↪ : L) * a7 * b6 = 0 →
  (9 : L) * a4 + (4 : L) * a5 * b9 + (-1 : L) * a6 * b8 + (-6 : L) * a7 * b7 = 0 →
  (6 : L) * a5 + (1 : L) * a6 * b9 + (-4 : L) * a7 * b8 = 0 →
  (3 : L) * a6 + (-2 : L) * a7 * b9 = 0 →
  a7 ^ 3 * b0 ^ 2 = 1 →
  ∃ w : L, w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0 ∧
  a1 = (((512502980) / 1428159681 : L) + ((8080) / 109858437 : L) * w +
  ↪ ((877688) / 1428159681 : L) * w ^ (2 : ℕ) + ((127760) / 84009393 : L) * w
  ↪ ^ (3 : ℕ) + ((-531068) / 1428159681 : L) * w ^ (4 : ℕ)) ∧
  a2 = (((7293893720809472) / 226626674937113529 : L) + ((285855137792) /
  ↪ 5810940383002911 : L) * w + ((-47390789046272) / 226626674937113529 : L) *
  ↪ w ^ (2 : ℕ) + ((-90487897751552) / 226626674937113529 : L) * w ^ (3 : ℕ) +
  ↪ ((21860765739008) / 226626674937113529 : L) * w ^ (4 : ℕ)) ∧
  a3 = (((646429323397524292763648) / 323659079784278752637424249 : L) +
  ↪ ((-311599571772213100544) / 24896852291098365587494173 : L) * w +
  ↪ ((-13107803800410363756544) / 323659079784278752637424249 : L) * w ^ (2 :
  ↪ ℕ) + ((-39360612513016248860672) / 323659079784278752637424249 : L) * w ^
  ↪ (3 : ℕ) + ((9886191189342570790912) / 323659079784278752637424249 : L) * w
  ↪ ^ (4 : ℕ)) ∧

```

```

a4 = (((4014890803348499913288661860352) /
↳ 154078949379156364060580574704501523 : L) +
↳ ((-19247583898558620695724556288) / 3950742291773240104117450633448757 :
↳ L) * w + ((-147813452969167985031874871296) /
↳ 154078949379156364060580574704501523 : L) * w ^ (2 : N) +
↳ ((-2132931317809853248718661222400) /
↳ 154078949379156364060580574704501523 : L) * w ^ (3 : N) +
↳ ((580398578162396326286055178240) / 154078949379156364060580574704501523
↳ : L) * w ^ (4 : N)) ^
a5 = (((-47195336258536808324667608668861104128) /
↳ 73349781064717033648626206081592521450564721 : L) +
↳ ((-2993278732025095090465120575366889472) /
↳ 5642290851132079511432785083199424726966517 : L) * w +
↳ ((3250009871460478519137256780580519936) /
↳ 8149975673857448183180689564621391272284969 : L) * w ^ (2 : N) +
↳ ((-35972900451489262851635484544208994304) /
↳ 73349781064717033648626206081592521450564721 : L) * w ^ (3 : N) +
↳ ((4212495817303089511647840819054903296) /
↳ 24449927021572344549542068693864173816854907 : L) * w ^ (4 : N)) ^
a6 = (((-35363777158142043633105970419962202288029696) /
↳ 24174276906186027491743446592090946632344039049203231 : L) +
↳ ((-24435887786194136948277618937091736054267904) /
↳ 1859559762014309807057188199391611279411079926861787 : L) * w +
↳ ((348725961520751776828947066063346243175710720) /
↳ 24174276906186027491743446592090946632344039049203231 : L) * w ^ (2 : N) +
↳ ((-8914342677872945930434815186362304837976064) /
↳ 1422016288599178087749614505417114507784943473482543 : L) * w ^ (3 : N) +
↳ ((67798985560990286741153010500606027934728192) /
↳ 24174276906186027491743446592090946632344039049203231 : L) * w ^ (4 : N))
↳ ^
a7 = (((342097029127255765817099097920847945090722114306048) /
↳ 885249425506264203827321815866695986293243258737477684336649 : L) +
↳ ((-5681054197220064630999879561744890228875177492480) /
↳ 22698703218109338559674918355556307340852391249678914982991 : L) * w +
↳ ((206885581446139393490715159158183040429422553333760) /
↳ 885249425506264203827321815866695986293243258737477684336649 : L) * w ^ (2
↳ : N) + ((-63348258959870509813954058121926485026619479031808) /
↳ 885249425506264203827321815866695986293243258737477684336649 : L) * w ^ (3
↳ : N) + ((25332256105915521578908408974616845866605607911424) /
↳ 885249425506264203827321815866695986293243258737477684336649 : L) * w ^ (4
↳ : N)) ^
b0 = (((241787651899027059649022111227674440235744649167926923464101195333242
↳ 8638485407) /
↳ 81144154259988381448527095356464165463993157634428720446762385408 : L) +
↳ ((1471147028669181503011061824254866913197610002216490166555765524465073
↳ 2041478549) /
↳ 81144154259988381448527095356464165463993157634428720446762385408 : L) *
↳ w + ((617854596280970291332161410119563350111260243577025705635188990137
↳ 876816776295) /
↳ 81144154259988381448527095356464165463993157634428720446762385408 : L) *
↳ w ^ (2 : N) + ((-36637857071993399425444586421057633386852605503275887942
↳ 21990329866702931989049) /
↳ 81144154259988381448527095356464165463993157634428720446762385408 : L) *
↳ w ^ (3 : N) + ((283919616043133221256996475572763279716168016664606915395
↳ 141666931558753453503) /
↳ 40572077129994190724263547678232082731996578817214360223381192704 : L) *
↳ w ^ (4 : N)) ^

```

```

b1 = (((213129345216263992177813311926195028654591343373099779726908014927974
↳ 58781) / 2064526619682179458796231817536743473030560696988314686717952 :
↳ L) + ((90786059190204626155643297641284354697393864232400683101193398378
↳ 854556479) /
↳ 2064526619682179458796231817536743473030560696988314686717952 : L) * w +
↳ ((3676839220908931005614147091295150358755034032737886207680394068928684
↳ 725) / 2064526619682179458796231817536743473030560696988314686717952 : L)
↳ * w ^ (2 : N) + ((-210875061122372142827023757868450079885049551692627677
↳ 71079750788915150635) /
↳ 2064526619682179458796231817536743473030560696988314686717952 : L) * w ^
↳ (3 : N) + ((1804471929942342272834364642391789791835513098528722294350400
↳ 675635340349) /
↳ 1032263309841089729398115908768371736515280348494157343358976 : L) * w ^
↳ (4 : N)) ^
b2 = (((-144544033917986925185258501166756369390602635518529573940467345287) /
↳ 410368263186114060193506031816247923444210651466039296 : L) +
↳ ((575311520507585192568556318243232964279467466281194477346330819301) /
↳ 102592065796528515048376507954061980861052662866509824 : L) * w +
↳ ((54330691268897368779422216815165957896195264760328594263494900495) /
↳ 205184131593057030096753015908123961722105325733019648 : L) * w ^ (2 : N)
↳ + ((-174187880018379336306198263901059240047180472649226203058128140895)
↳ / 102592065796528515048376507954061980861052662866509824 : L) * w ^ (3 :
↳ N) + ((80166557140337989909132856542161474521909286203155803758626110033)
↳ / 410368263186114060193506031816247923444210651466039296 : L) * w ^ (4 :
↳ N)) ^
b3 = (((-7425458405034303101010032795601866242474210402999635943131) /
↳ 20392339559214813347634883685912610051826122752 : L) +
↳ ((6744315542914009035390981237841878170110250243622814128087) /
↳ 20392339559214813347634883685912610051826122752 : L) * w +
↳ ((478440749235744236134872881006436261827393493416098817853) /
↳ 20392339559214813347634883685912610051826122752 : L) * w ^ (2 : N) +
↳ ((-3700344121490310017718636169454092008434178658425433227171) /
↳ 20392339559214813347634883685912610051826122752 : L) * w ^ (3 : N) +
↳ ((6223309793893257222702608499502859806885134430639402117) /
↳ 10196169779607406673817441842956305025913061376 : L) * w ^ (4 : N)) ^
b4 = (((-196982124334537966834877980654129755738767580317247) /
↳ 4053408121472769419865599653882359709696 : L) +
↳ ((7356162939037843464175425549716484499362290432253) /
↳ 1013352030368192354966399913470589927424 : L) * w +
↳ ((3334214783216888283093389346861252319817155285863) /
↳ 2026704060736384709932799826941179854848 : L) * w ^ (2 : N) +
↳ ((-13592771536859666166850556543437368777444921613959) /
↳ 1013352030368192354966399913470589927424 : L) * w ^ (3 : N) +
↳ ((-1743167522426431101053174423220977563711561703271) /
↳ 4053408121472769419865599653882359709696 : L) * w ^ (4 : N)) ^
b5 = (((-5945326279911459300427967186228236954680873) /
↳ 1611400923519540560385711240380416 : L) +
↳ ((-1207414796044598637778246641365546517002499) /
↳ 1611400923519540560385711240380416 : L) * w +
↳ ((147397906832220690705397488682848525927903) /
↳ 1611400923519540560385711240380416 : L) * w ^ (2 : N) +
↳ ((-1003046473955460175839195938091594613306305) /
↳ 1611400923519540560385711240380416 : L) * w ^ (3 : N) +
↳ ((-57989457976321180426899430535315371961481) /
↳ 805700461759770280192855620190208 : L) * w ^ (4 : N)) ^
b6 = (((-40717564079347560855720292779927573) / 320299962217494673010720768 :
↳ L) + ((-4431956341187962499826675611271825) / 80074990554373668252680192 :
↳ L) * w + ((755747337967616823449366320710957) /
↳ 160149981108747336505360384 : L) * w ^ (2 : N) +
↳ ((-869471274892038676474107514591197) / 80074990554373668252680192 : L) *
↳ w ^ (3 : N) + ((-93883857777777016547324128345997) /
↳ 320299962217494673010720768 : L) * w ^ (4 : N)) ^

```

```

b7 = (((-42258994952000509880862795) / 31833190703543877632 : L) +
  ↪ ((-36376188014455196253194745) / 31833190703543877632 : L) * w +
  ↪ ((5527319619223464764982093) / 31833190703543877632 : L) * w ^ (2 : ℕ) +
  ↪ ((3622463246490253699729005) / 31833190703543877632 : L) * w ^ (3 : ℕ) +
  ↪ ((-717039634870611328721643) / 15916595351771938816 : L) * w ^ (4 : ℕ)) ^
b8 = (((89529406222349457) / 6327518887936 : L) + ((-9772922162070171) /
  ↪ 1581879721984 : L) * w + ((8308630462897287) / 3163759443968 : L) * w ^ (2
  ↪ : ℕ) + ((8039403284956497) / 1581879721984 : L) * w ^ (3 : ℕ) +
  ↪ ((-2811959945010567) / 6327518887936 : L) * w ^ (4 : ℕ)) ^
b9 = (((1461767229) / 5030912 : L) + ((82735263) / 5030912 : L) * w +
  ↪ ((34254981) / 5030912 : L) * w ^ (2 : ℕ) + ((182339829) / 5030912 : L) * w
  ↪ ^ (3 : ℕ) + ((-10377315) / 2515456 : L) * w ^ (4 : ℕ))

```

The top-layer classification statement and its witnesses:

Listing 13: Mechanically extracted from Jacobian/BranchAbClassification.lean (sha256 29fdec5a9fda336); definition printed verbatim in full.

```

def TopLayerClassification (L : Type*) [Field L] : Prop :=
  ∀ (A₂ B₃ : L[X]) (lam : L),
    (∀ i, A₂.coeff i ≠ 0 → 1 ≤ i ∧ i ≤ 8) → (∀ i, B₃.coeff i ≠ 0 → 2 ≤ i ∧ i ≤ 12) →
    A₂.coeff 1 ≠ 0 → A₂.coeff 8 ≠ 0 → B₃.coeff 2 ≠ 0 → B₃.coeff 12 ≠ 0 →
    layerTerm 2 3 A₂ B₃ = C lam * X ^ 2 →
    ∃ w ρ σ ε : L, w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0 ∧ ρ ≠ 0 ∧ σ ≠ 0 ∧ ε
    ↪ ≠ 0 ∧
    (∀ i, A₂.coeff i = ρ * ε ^ (i - 1) * topA w i) ∧ (∀ k, B₃.coeff k = σ * ε ^ (k
    ↪ - 2) * topB w k)

```

Listing 14: Mechanically extracted from Jacobian/BranchAbSharp.lean (sha256 9eb4eb229d2047f6); definition printed verbatim in full.

```

noncomputable def topA {L : Type*} [Field L] (w : L) : ℕ → L
| 1 => ((1 : L))
| 2 => (((303) / 16 : L) + ((879) / 128 : L) * w + ((-389) / 128 : L) * w ^ (2 : ℕ)
  ↪ + ((-51) / 128 : L) * w ^ (3 : ℕ) + ((49) / 128 : L) * w ^ (4 : ℕ))
| 3 => (((315) / 16 : L) + ((693) / 8 : L) * w + ((-147) / 8 : L) * w ^ (2 : ℕ) +
  ↪ ((-189) / 8 : L) * w ^ (3 : ℕ) + ((147) / 16 : L) * w ^ (4 : ℕ))
| 4 => (((-79653) / 64 : L) + ((29463) / 64 : L) * w + ((7371) / 128 : L) * w ^ (2
  ↪ : ℕ) + ((-4767) / 16 : L) * w ^ (3 : ℕ) + ((8001) / 128 : L) * w ^ (4 : ℕ))
| 5 => (((-33519) / 4 : L) + ((-6843) / 4 : L) * w + ((8181) / 8 : L) * w ^ (2 : ℕ)
  ↪ + ((-1713) / 2 : L) * w ^ (3 : ℕ) + ((-1305) / 8 : L) * w ^ (4 : ℕ))
| 6 => (((-8805) / 8 : L) + ((-76965) / 8 : L) * w + ((11523) / 16 : L) * w ^ (2 :
  ↪ ℕ) + ((6927) / 4 : L) * w ^ (3 : ℕ) + ((-18579) / 16 : L) * w ^ (4 : ℕ))
| 7 => (((389740) / 17 : L) + ((-51922) / 17 : L) * w + ((-48572) / 17 : L) * w ^
  ↪ (2 : ℕ) + ((63239) / 17 : L) * w ^ (3 : ℕ) + ((-6475) / 17 : L) * w ^ (4 : ℕ))
| 8 => (((476437) / 68 : L) + ((314405) / 68 : L) * w + ((-174571) / 136 : L) * w ^
  ↪ (2 : ℕ) + ((1574) / 17 : L) * w ^ (3 : ℕ) + ((82783) / 136 : L) * w ^ (4 : ℕ))
| _ => 0

```

Listing 15: Mechanically extracted from Jacobian/BranchAbSharp.lean (sha256 9eb4eb229d2047f6); definition printed verbatim in full.

```

noncomputable def topB {L : Type*} [Field L] (w : L) : ℕ → L
| 2 => ((1 : L))

```

```

| 3 => (((101) / 8 : L) + ((293) / 64 : L) * w + ((-389) / 192 : L) * w ^ (2 : N) +
  ↪ ((-17) / 64 : L) * w ^ (3 : N) + ((49) / 192 : L) * w ^ (4 : N))
| 4 => (((315) / 16 : L) + ((693) / 8 : L) * w + ((-147) / 8 : L) * w ^ (2 : N) +
  ↪ ((-189) / 8 : L) * w ^ (3 : N) + ((147) / 16 : L) * w ^ (4 : N))
| 5 => (((-11379) / 8 : L) + ((4209) / 8 : L) * w + ((1053) / 16 : L) * w ^ (2 : N)
  ↪ + ((-681) / 2 : L) * w ^ (3 : N) + ((1143) / 16 : L) * w ^ (4 : N))
| 6 => (((-695299) / 48 : L) + ((-48367) / 48 : L) * w + ((52679) / 32 : L) * w ^
  ↪ (2 : N) + ((-45661) / 24 : L) * w ^ (3 : N) + ((-2211) / 32 : L) * w ^ (4 : N))
| 7 => (((-309895) / 8 : L) + ((-251275) / 8 : L) * w + ((101481) / 16 : L) * w ^
  ↪ (2 : N) + ((1811) / 4 : L) * w ^ (3 : N) + ((-60489) / 16 : L) * w ^ (4 : N))
| 8 => (((1733348) / 13 : L) + ((-2368725) / 26 : L) * w + ((-184976) / 13 : L) * w
  ↪ ^ (2 : N) + ((1819939) / 52 : L) * w ^ (3 : N) + ((-569071) / 52 : L) * w ^ (4
  ↪ : N))
| 9 => (((18736247) / 39 : L) + ((1087133) / 13 : L) * w + ((-5656777) / 78 : L) *
  ↪ w ^ (2 : N) + ((1993384) / 39 : L) * w ^ (3 : N) + ((971365) / 78 : L) * w ^ (4
  ↪ : N))
| 10 => (((305098) / 17 : L) + ((5784066) / 17 : L) * w + ((-308335) / 17 : L) * w
  ↪ ^ (2 : N) + ((-950608) / 17 : L) * w ^ (3 : N) + ((754021) / 17 : L) * w ^ (4 :
  ↪ N))
| 11 => (((-149721182) / 323 : L) + ((26391690) / 323 : L) * w + ((21560033) / 323
  ↪ : L) * w ^ (2 : N) + ((-24378412) / 323 : L) * w ^ (3 : N) + ((2786935) / 323 :
  ↪ L) * w ^ (4 : N))
| 12 => (((-91912444) / 969 : L) + ((-20715500) / 323 : L) * w + ((17474146) / 969
  ↪ : L) * w ^ (2 : N) + ((-1623428) / 969 : L) * w ^ (3 : N) + ((-8783302) / 969 :
  ↪ L) * w ^ (4 : N))
| _ => 0

```

6.3 Irreducibility and reality

Theorem 6.3 (Top-layer structure). *Let $\mathcal{W} \in \mathbb{Z}[a_6]$ be the degree-35 eliminating polynomial for a_6 at $m = 7$. Then:*

- (i) $\mathcal{W}$ is irreducible over $\mathbb{Q}$ (SINGULAR factorize over $\mathbb{Q}$ yields a nonzero constant times a single degree-35 factor);
- (ii) $\mathcal{W}(a_6) = V(a_6^7)$ for a quintic $V \in \mathbb{Z}[S]$;
- (iii) V is separable (nonzero discriminant) and has exactly one real root, $S \approx 3208.612478738048$;
- (iv) $\mathcal{W}$ is separable modulo 101 ($\gcd(\mathcal{W}, \mathcal{W}') = 1$ in $\mathbb{F}_{101}[a_6]$), so the 35 complex solutions are distinct.

Consequently the 35 solutions form a single Galois orbit over $\mathbb{Q}$, comprising one real solution and 17 complex-conjugate pairs. See Figure 3.

Proof. (i) Direct factorization in SINGULAR over $\mathbb{Q}$. (ii) By inspection: every exponent of $\mathcal{W}$ is divisible by 7. (iii) Discriminant computed exactly in SYMPY (nonzero integer); real-root count via Sturm sequence gives exactly one real S . (iv) Direct gcd computation in $\mathbb{F}_{101}[a_6]$. $\square$

Remark 6.4 (Characteristic-zero/modular consistency). Reducing the integer quintic V modulo 101 gives 58 times the quintic obtained directly from the mod-101 Gröbner computation (equivalently, the mod-101 quintic is 54 times V), confirming consistency between the characteristic-zero and modular routes.

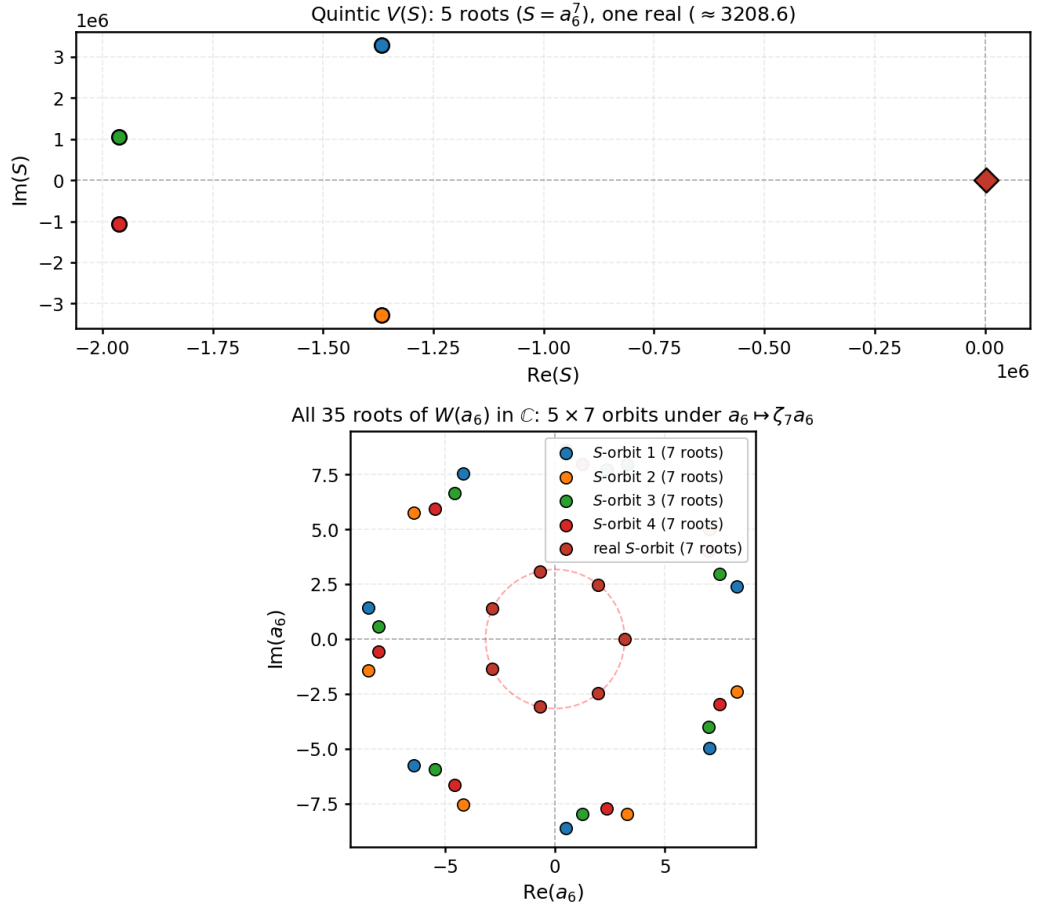


Figure 3: Top: the five roots of the quintic $V(S)$ in the complex S -plane, computed from the integer coefficients; the real root $S \approx 3208.61$ is marked with a diamond. Bottom: all 35 roots of $W(a_6)$ in $\mathbb{C}$, colored by S -orbit; each $S \neq 0$ lifts to 7 values of a_6 via 7th roots of unity. The dashed circle marks the real S -orbit. The 35 roots form a single Galois orbit over $\mathbb{Q}$.

6.4 An algebraic proof of the top-layer classification for $m = 7$, and its formalization

The proof of Proposition 6.1 above rests on Riemann's existence theorem, which is not in Mathlib (at least not in the version pinned by the artifact). For $m = 7$ we now give a second proof of the classification it provides, in the normalized chart of §5: every solution is one of the points $(P_j(w), Q_k(w))$ with $R(w) = 0$, the five K_5 -conjugates. This is the form in which the main theorem uses Proposition 6.1. The eliminant $\mathcal{W}$ and the cases $m = 3, 5$ are not reproved here. The proof is purely algebraic and valid over every field of characteristic 0, and it is formalized: Theorem 6.5 is the Lean theorem `chartClassification_holds` (listing 16). We work in the Lean chart of §5: $\alpha = 1 + a_1u + \dots + a_7u^7$, $\beta = b_0 + b_1u + \dots + b_9u^9 + u^{10}$, and $E_n = \sum_{i+k=n} (1 + 2k - 3i) a_i b_k$ ($1 \leq n \leq 16$) is the coefficient of u^n in $\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta)$.

Theorem 6.5 (`chartClassification_holds`). *Let L be a field of characteristic 0. Suppose $(a_1, \dots, a_7, b_0, \dots, b_9) \in L^{17}$ satisfies $E_1 = \dots = E_{16} = 0$ and $a_7^3 b_0^2 = 1$. Then $(a_j, b_k) = (P_j(w), Q_k(w))$ for a root $w \in L$ of R . Here $P_j, Q_k \in \mathbb{Q}[w]$ are the coordinates in listing 12.*

Proof. Step 1: reversed coordinates. Since $a_7^3 b_0^2 = 1$, we have $a_7 \neq 0$. Put $y_i = a_{7-i}/a_7$ for $1 \leq i \leq 6$ and $y_7 = 1/a_7$, and give y_i the weight i . Define $S_k \in \mathbb{Q}[y_1, \dots, y_7]$, weighted homogeneous of weight k , by

$$(1 + y_1 v + \dots + y_7 v^7)^{3/2} = \sum_{k \geq 0} S_k v^k,$$

or, equivalently, by $S_0 = 1$ and $2k S_k = \sum_{i=1}^7 (5i - 2k) y_i S_{k-i}$. Write $\tilde{\beta}_k = b_{10-k}$ for $0 \leq k \leq 10$ (so $\tilde{\beta}_0 = 1$) and $\tilde{\beta}_k = 0$ for $k > 10$; in both recursions, terms with a negative index are zero. Reindexing E_{17-k} by $i \mapsto 7 - i$, $k \mapsto 10 - k$ gives

$$E_{17-k}/a_7 = -2k \tilde{\beta}_k + \sum_{i=1}^7 (5i - 2k) y_i \tilde{\beta}_{k-i} \quad (1 \leq k \leq 16).$$

This is the recursion for S_k . Induction on k gives $b_{10-k} = S_k(y)$ for $1 \leq k \leq 10$, and

$$S_{11}(y) = \dots = S_{16}(y) = 0, \quad S_{10}(y)^2 = y_7^3. \quad (7)$$

The last equation holds because $b_0 = S_{10}(y)$ and $a_7^3 b_0^2 = 1$. In the language of §6, the reversed β is the truncation of the power-series square root of the reversed α^3/a_7^3 . This is the Davenport–Stothers condition.

Step 2: the orbit relations. Consider the six weighted-homogeneous relations

$$\begin{aligned} y_4 &= \frac{2431}{18240} y_1^4 - \frac{1027}{1710} y_1^2 y_2 + \frac{143}{380} y_2^2 + \frac{41}{60} y_1 y_3, \\ y_5 &= \frac{1154351}{25313472} y_1^5 - \frac{3833107}{18985104} y_1^3 y_2 + \frac{6512}{43947} y_1 y_2^2 + \frac{24449}{166536} y_1^2 y_3, \\ y_6 &= \frac{4206749569}{4242031637760} y_1^6 + \frac{8589189473}{1590761864160} y_1^4 y_2 - \frac{440013827}{9819517680} y_1^2 y_2^2 \\ &\quad + \frac{7454351}{734423760} y_1^3 y_3 + \frac{253}{5054} y_2^3, \\ y_7 &= -\frac{4539711243127}{16353031963564800} y_1^7 + \frac{13373800868041}{6132386986336800} y_1^5 y_2 - \frac{214713769679}{37854240656400} y_1^3 y_2^2 \\ &\quad + \frac{81661753}{707800898700} y_1^4 y_3 + \frac{92333}{19483170} y_1 y_2^3, \\ y_2 y_3 &= \frac{3738317}{46630080} y_1^5 - \frac{16882087}{34972560} y_1^3 y_2 + \frac{115021}{161910} y_1 y_2^2 + \frac{2277881}{5828760} y_1^2 y_3, \\ y_3^2 &= \frac{740761013029}{10605079094400} y_1^6 - \frac{754440973541}{1988452330200} y_1^4 y_2 + \frac{3827878261}{8182931400} y_1^2 y_2^2 \\ &\quad + \frac{137816863}{918029700} y_1^3 y_3 + \frac{289}{12635} y_2^3. \end{aligned}$$

Write $g = 0$ for any of them. Each satisfies

$$y_7^3 \cdot g \in (S_{11}, \dots, S_{16}) \subset \mathbb{Q}[y_1, \dots, y_7],$$

with explicit cofactors of 341 to 694 terms, whose numerators and denominators have at most 42 digits. Since $y_7 \neq 0$, every solution of (7) satisfies all six relations.

The relations were found by exact interpolation on the K_5 point, as all weighted-homogeneous forms of weight at most 7 that vanish on the five orbits. The cofactors come from exact weighted-homogeneous linear algebra over $\mathbb{Q}$. Let (g) be the ideal of the six relations. A Gröbner computation over $\mathbb{Q}$ (`saturation_check.py`) shows that $(S_{11}, \dots, S_{16}) \subseteq (g)$ and $(g) : y_7 = (g)$. With the certificates this gives

$$(S_{11}, \dots, S_{16}) : y_7^\infty = (S_{11}, \dots, S_{16}) : y_7^3 = (g).$$

The ideal (g) has Krull dimension 1 and needs all six generators, so it is a weighted complete intersection. It has exactly five solutions in the chart $y_1 = 1$ and none with $y_1 = 0$ other than $y = 0$, in agreement with weighted Bézout, $4 \cdot 5 \cdot 5 \cdot 6 \cdot 6 \cdot 7/7! = 5$. These facts explain the choice of relations but are not used in the proof.

Step 3: a resultant. The relation for y_7 shows $y_7 \in y_1 \mathbb{Q}[y_1, y_2, y_3]$, so $y_1 \neq 0$. Put $t = y_2/y_1^2$ and $s = y_3/y_1^3$. The relation for $y_2 y_3$ becomes $(t - c)s = N(t)$ with $c = \frac{2277881}{5828760}$ and $\deg N = 2$, and the relation for y_3^2 becomes monic quadratic in s . Their Sylvester resultant in s is $\kappa m(t)$ with $\kappa \in \mathbb{Q}^\times$ and

$$\begin{aligned} m(t) = & 287548593020928 t^5 - 688401965085696 t^4 + 640652914818432 t^3 \\ & - 292066554895024 t^2 + 65563255857792 t - 5817852446211, \end{aligned}$$

which is irreducible over $\mathbb{Q}$. Hence $t - c$ is invertible modulo m , and $s = q_3(t)$. The first four relations then give $y_i/y_1^i = q_i(t)$ for $4 \leq i \leq 7$, with $q_i \in \mathbb{Q}[t]$ reduced modulo m .

Step 4: the torus. Let $z = 1/y_1$. By weighted homogeneity,

$$z^{10} S_{10}(y) = S_{10}(1, t, q_3(t), \dots, q_7(t)) \equiv \sigma(t) \pmod{m} \quad \text{and} \quad z^7 y_7 = q_7(t).$$

Multiplying $S_{10}^2 = y_7^3$ by z^{21} gives $z \sigma(t)^2 = q_7(t)^3$, that is,

$$y_1 = \sigma(t)^2 q_7(t)^{-3} \pmod{m(t)}.$$

This fixes the torus without a seventh root, as in §5, and makes every y_i a polynomial in t modulo m . The substitution $w = \omega(t)$, with $\omega \in \mathbb{Q}[t]$ of degree 4, induces an isomorphism $\mathbb{Q}[t]/(m) \cong \mathbb{Q}[w]/(R) = K_5$ and gives $R(w) = 0$ and $y_i = Y_i(w)$.

Step 5: back to the chart. The formulas $a_7 = 1/y_7$ and $a_{7-i} = y_i/y_7$, together with the recursion of Step 1 for b_{10-k} , give $a_j = P_j(w)$ and $b_k = Q_k(w)$. $\square$

Remark 6.6 (Formalization). The proof is formalized in `lean/Jacobian/ChartProof/`: 12 modules, all generated by `lean/certgen/chartproof/` except the hand-written `Reflect.lean`.

- **Steps.** There are 111 steps. In 105 of them a polynomial g vanishes because $g = \sum_i c_i e_i$ for polynomials e_i already known to vanish. In the other six, one for each relation of Step 2, the identity $y_7^3 g = e$ holds with e already known to vanish, and y_7^3 is cancelled using $y_7 \neq 0$.
- **Kernel checking.** Every identity is checked by the Lean kernel. `lc_zero` (listing 18) and `eq_of_toPolyK` reduce it to an equality of normal forms. These are computed by the verified normalizer of `Lean.Grind.CommRing` from Lean core, with a kernel-friendly multiplication defined by recursors (`toPolyK`, soundness theorem `denote_toPolyK`), and `decide +kernel` evaluates them.
- **Axioms.** No `native_decide` is used. `#print axioms` for `chartClassification_holds` and for `main_theorem` reports only `propext`, `Classical.choice` and `Quot.sound`.

- *Batching.* The certificates of Step 2 have between 27,747 and 56,999 monomial products each. They are split into batches of at most 9,000 products, one declaration per batch.
- *Independent checks.* Every certificate was also verified by exact SymPy expansion before it was written, and the generator reproduces all eleven generated modules byte-identically. A second checker, written independently of the generator (independent_check.py), parses the Lean text and re-verifies all 111 identities over $\mathbb{Z}$. Six controls behave as expected: four one-integer perturbations are rejected by decide, a planted sorry is flagged, and an unmodified module is accepted.
- *Cost.* The build takes about 25 minutes sequentially and needs at most 5.4 GB of memory for any one module.

Remark 6.7 (Why these coordinates). Earlier formulations failed to yield certificates of practical size because of the degenerate Davenport–Stothers components of lower degree (the solutions for $m = 1, 3, 5$). With $a_0 = b_0 = 1$ and b eliminated, these components force saturation by a_7^7 or a_8^8 , so the certificates have weight 61. In the chart $a_0 = b_{10} = 1$, the sub-case $a_1 = 0$ alone needed 503,254 cofactor terms modulo a prime. Replayed over $\mathbb{Q}$, its Buchberger trace reached coefficients of 2,722 digits after 929 of its 2,065 elements (lean/CLASSIFICATION_STATUS.md, §5). In the reversed coordinates the same components lie on $y_7 = 0$, and y_7^3 suffices, at weight at most 28. The five orbits then form a weighted complete intersection that can be written down explicitly. The six certificates of Step 2 have 341 to 694 cofactor terms each over $\mathbb{Q}$, roughly a thousand times fewer than the modular lift for $a_1 = 0$ alone.

Listing 16: Mechanically extracted from Jacobian/ChartProof/Final.lean (sha256 638b644ffb58cb8b); statement verbatim, proof elided.

```
theorem chartClassification_holds (L : Type*) [Field L] [CharZero L] :
  ⇐ ChartClassification L := by
  -- proof body elided (see artifact)
```

Listing 17: Mechanically extracted from Jacobian/ChartProof/Final.lean (sha256 638b644ffb58cb8b); statement verbatim, proof elided.

```
theorem main_theorem {L : Type*} [Field L] [CharZero L] :
  ¬ ∃ (P Q : MvPolynomial (Fin 2) L) (lam : L), lam ≠ 0 ∧ NewtonNF2 P Q ∧ jac P Q =
  ⇐ MvPolynomial.C lam * MvPolynomial.X 0 ^ 2 :=
  -- proof term elided (see artifact)
```

Listing 18: Mechanically extracted from Jacobian/ChartProof/Reflect.lean (sha256 9e3ea428ffe7236b); statement verbatim, proof elided.

```
theorem lc_zero (ctx : Context α) (g : Expr) (l : List (Expr × Expr))
  (hcert : Poly.beq' (toPolyK (.sub g (lcExpr l))) (.num 0) = true) (h : AllZero
  ⇐ ctx l) :
  g.denote ctx = 0 := by
  -- proof body elided (see artifact)
```

7 Descent

Fix a top-layer solution (α, β) . The descent solves E_4, E_3, E_2 in turn. We use the scaled variables $A_2 = u\alpha$ and $B_3 = u^2\beta$.

7.1 Layer E_4

Proposition 7.1 (E_4 structure). *E_4 is a linear system in 19 unknowns (A_1 on $u^1, \dots, u^8$; B_2 on $u^2, \dots, u^{12}$) with 18 equations, rank 17, and kernel dimension 2. The kernel is parametrized by $t = (t_1, t_2)$, giving $(A_1(t), B_2(t))$ linear in t . Computed exactly in characteristic zero over the quintic field $K_5 = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$ (script `exact_ranks_K5.py`); the modular evaluation at $p = 101$, $a_6 = 55$ is an independent cross-check.*

Proof. Rank computation on the 18×19 coefficient matrix, with residual asserts verifying the kernel. Scripts: `exact_ranks_K5.py` (exact rank and kernel over K_5) and `C5_E2_obstruction_corrected.py` (which verifies E_4 with E_4 -kernel residuals zero; mod-101 cross-check). $\square$

7.2 Layer E_3

Proposition 7.2 (E_3 structure and unconditional solvability). *E_3 is a 19×20 linear system of rank 18 with 2-dimensional kernel. Moreover, E_3 is solvable for every direction $t = (t_1, t_2)$: writing the system as $M(t)x = K(t)$ with M constant and K quadratic in t , the left nullvector v (with $v^\top M = 0$) satisfies*

$$v^\top K_{11} \equiv 0, \quad v^\top K_{22} \equiv 0, \quad v^\top K_{12} \equiv 0 \quad (8)$$

identically as polynomials in t , where $K = t_1^2 K_{11} + t_1 t_2 K_{12} + t_2^2 K_{22}$. Hence no direction is obstructed at E_3 . All computations are exact over K_5 ; see Lemma 9.1(i).

Proof. The 19×20 matrix M has rank 18; its left nullspace is 1-dimensional, spanned by v . Decomposing the inhomogeneous part and computing $v^\top K_{ij}$ symbolically gives zero for all i, j . Script: `C5_E2_obstruction_corrected.py` (which verifies E_3 with residual asserts). $\square$

The general solution of E_3 is thus the 2-parameter affine family

$$x(t, s) = t_1^2 x_{11} + t_1 t_2 x_{12} + t_2^2 x_{22} + s_1 w_1 + s_2 w_2, \quad (9)$$

encoding (A_0, B_1) in terms of $(t, s) = (t_1, t_2, s_1, s_2) \in \mathbb{C}^4$.

7.3 Layer E_2

Proposition 7.3 (E_2 injectivity). *The homogeneous operator of E_2 in the unknown B'_0 is a 19×12 matrix of rank 12, hence injective. Its left nullspace has dimension $19 - 12 = 7$. Exact over K_5 ; see Lemma 9.1(i)–(ii).*

Therefore B'_0 (and hence B_0 up to an additive constant) is uniquely determined by (A_0, B_1, A_1, B_2) . The seven left-nullvector pairings then give seven compatibility conditions:

$$W \cdot \text{rhs}(t, s) = 0, \quad (10)$$

where W is 7×19 and $\text{rhs}(t, s)$ is the E_2 inhomogeneous term. These seven conditions are the *only* remaining obstructions; E_1 is never needed.

Listing 19: Mechanically extracted from Jacobian/Descent/Main.lean (sha256 e18a3a9e531df6d6); statement verbatim, proof elided.

```
theorem descent_K5 {L : Type*} [Field L] [CharZero L]
  (w : L)
  (hw : w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0)
  (a_1_0 a_2_2 a_3_4 a_4_6 a_5_8 a_6_10 a_7_12 a_8_14 b_2_1 b_3_3 b_4_5 b_5_7
   ↪ b_6_9 b_7_11 b_8_13 b_9_15 b_10_17 b_11_19 b_12_21 : L)
  (h_a_1_0 : a_1_0 = ((1 : L)))
  -- ... (18 h4_*, 19 h3_*, 19 h2_*, 8 h_a_*, 11 h_b_*, 51 lower-layer unknowns in
   ↪ one binder)
  (hv : b_12_24 ≠ 0) :
  False := by
  -- proof body elided (see artifact)
```

8 The final obstruction

We use the homogeneous formulation: E_3 with A_0 on $u^1, \dots, u^8$ ($a_{8,16}$ free, no constant term) and B_1 on $u^1, \dots, u^{12}$. This gives the genuine 2-dimensional kernel with two independent directions $s = (s_1, s_2)$, both moving actual coefficients. (An earlier formulation froze $a_{8,16} = 1$ and included the constant term as an unknown; the constant is killed by A'_0 and contributes only a trivial direction.)

Remark 8.1 (Free parameter). §5 leaves $a_{8,16}$ free from the start. The E_2/E_3 analysis therefore applies to every slice, including $a_{8,16} = 1$, without a superset argument.

8.1 Explicit form of the seven conditions

Expanding (10) symbolically in (t_1, t_2, s_1, s_2) gives:

Lemma 8.2. *Each of the seven conditions has the form*

$$b_i(t_1, t_2) + s_1 \cdot M_i(t_1, t_2) + s_2 \cdot L_i(t_1, t_2) = 0, \quad i = 1, \dots, 7, \quad (11)$$

where b_i is homogeneous cubic and M_i, L_i are linear in (t_1, t_2) . Both s_1 and s_2 appear genuinely.

Proof. Exact expansion over K_5 in python-flint (script `exact_obstruction_K5.py`) of $W \cdot \text{rhs}(t, s)$ with the homogeneous E_3 parametrization $x(t, s) = t_1^2 x_{11} + t_1 t_2 x_{12} + t_2^2 x_{22} + s_1 w_1 + s_2 w_2$ substituted; the SYMPY expansion over $\mathbb{F}_{101}$ (script `C5_E2_obstruction_corrected.py`) is a cross-check. $\square$

8.2 Elimination of (s_1, s_2)

For fixed $t \neq 0$, the seven equations are affine-linear in (s_1, s_2) . A solution can exist only if the 7×3 matrix $[b(t) \mid M(t) \mid L(t)]$ has rank at most 2, i.e. only if all 3×3 minors vanish:

$$\det \begin{pmatrix} b_i & M_i & L_i \\ b_j & M_j & L_j \\ b_k & M_k & L_k \end{pmatrix} = 0, \quad 1 \leq i < j < k \leq 7. \quad (12)$$

There are $\binom{7}{3} = 35$ such minors, each a homogeneous binary quintic in (t_1, t_2) (cubic $\times$ linear $\times$ linear).

Theorem 8.3 (No nonzero direction). *The 35 quintics (12) have Gröbner basis*

$$(t_1, t_2)^5 = \{ t_1^5, t_1^4 t_2, t_1^3 t_2^2, t_1^2 t_2^3, t_1 t_2^4, t_2^5 \} \quad (13)$$

exactly over K_5 (the 35×6 coefficient matrix has rank 6, spanning all binary quintics). The mod-101 Gröbner computation (lex order) is an independent cross-check. Hence their only common zero in $\overline{K_5}^2$ is $(0, 0)$: there is no nonzero t admitting (s_1, s_2) satisfying the seven E_2 conditions.

Proof. Exact computation over K_5 : the 35 nonzero 3×3 minors span all binary quintics (rank 6), generating $(t_1, t_2)^5$. The radical is (t_1, t_2) , so the affine variety over the algebraic closure is $\{(0, 0)\}$. Script: `exact_obstruction_K5.py`. The SYMPY Gröbner basis over $GF(101)$ (script `C5_E2_obstruction_corrected.py`, saved, rerunnable; log archived) is a cross-check. $\square$

8.3 The case $t = 0$

Corollary 8.4 ($t = 0$ impossible). *Because the target bracket identity $[P, Q] = x^2 = u^2 y^{-4}$ contains only a y^{-4} component, the layer equation E_4 at y^{-3} is strictly homogeneous:*

$$2(A_2 B'_2 - A'_2 B_2) + (A_1 B'_3 - 3A'_1 B_3) = 0.$$

Consequently, setting the kernel parameter $t = (t_1, t_2) = (0, 0)$ forces $(A_1, B_2) \equiv (0, 0)$. In layer E_3 , the quadratic driving term $K(t)$ vanishes identically at $t = 0$, leaving only the homogeneous kernel $(A_0, B_1) = s_1 w_1 + s_2 w_2$. In layer E_2 , the terms $(A_1 B'_1 - A'_1 B_1) - 2A'_0 B_2$ vanish identically, reducing the compatibility equation strictly to $2A_2 B'_0 = 0$. Since $A_2 = u\alpha \neq 0$, this requires $B'_0 = 0$, forcing $B_0(u)$ to be constant. Hence $b_{12,24} = 0$; in particular, $(12, 24)$ cannot be a vertex of $N(Q)$.

Remark 8.5. The Lean formalization `e2_t0` (listing 20) proves $b_{12,24} = 0$ via a *different route*: a left-inverse certificate on the reduced E_2 equations with $t = 0$ supplied as hypotheses, rather than the $2A_2 B'_0 = 0$ argument above. Both routes reach the same conclusion.

Listing 20: Mechanically extracted from Jacobian/Descent/E2/T0.lean (sha256 3559bcdbbff42d3c); statement verbatim, proof elided.

```
theorem e2_t0 {L : Type*} [Field L] [CharZero L]
  (w : L)
  (hw : w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0)
  (b_11_20 b_12_22 b_11_21 b_12_23 b_1_2 b_2_4 b_3_6 b_4_8 b_5_10 b_6_12 b_7_14
   ↪ b_8_16 b_9_18 b_10_20 b_11_22 b_12_24 : L)
  -- 19 reduced E2 equations (r2_1_1) ... (r2_19_37), hypotheses elided for brevity
  (ht1 : b_11_20 = 0)
  (ht2 : b_12_22 = 0) :
  b_12_24 = 0 := by
  -- proof body elided (see artifact)
```

8.4 Proof of the main theorem

Proof of Theorem 1.1. By Proposition 4.1, the layers satisfy $E_5, \dots, E_1$. Since $a_{1,0} \cdot a_{8,14} \cdot b_{2,1} \cdot b_{12,21} \neq 0$, Proposition 6.1 puts (A_2, B_3) in the torus orbit of the K_5 point for some root w of R . By torus transport (Listing 6) we may take it to be that point. Propositions 7.1–7.3 and Theorem 8.3 force $t = 0$. At $t = 0$, Corollary 8.4’s computation gives $B'_0 = 0$, so $b_{12,24} = 0$ (in Lean, `e2_t0`, Listing 20). `a816_full.sing` (`scripts/a816_full.sing`) gives $a_{8,16} = 0$ at the K_5 point once E_1 is included. Torus transport multiplies $a_{8,16}$ by a nonzero scalar, so $a_{8,16} = 0$ on the whole orbit. The last sentence of the theorem follows because `NewtonNF2` requires $b_{12,24} \neq 0$. $\square$

Proof of Corollary 1.2. Proposition 4.3 of [1], then Theorem 1.1. $\square$

Listing 21: Mechanically extracted from `Jacobian/BranchAbFinal.lean` (`sha256 cd4e207adba5cb7b`); statement verbatim, proof elided.

```
theorem no_completion_K5 {L : Type*} [Field L] [CharZero L] (w : L)
  (hw : w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0) (lam : L)
  (A0 A1 A2 B0 B1 B2 B3 : Polynomial L) (p0 p1 p2 q0 q1 q2 q3 : MvPolynomial (Fin
    ↪ 2) L)
  (hp0 : X 1 ^ 0 * p0 = evH A0) (hp1 : X 1 ^ 1 * p1 = evH A1) (hp2 : X 1 ^ 2 * p2 =
    ↪ evH A2)
  (hq0 : X 1 ^ 0 * q0 = evH B0) (hq1 : X 1 ^ 1 * q1 = evH B1) (hq2 : X 1 ^ 2 * q2 =
    ↪ evH B2)
  (hq3 : X 1 ^ 3 * q3 = evH B3)
  (hJ : jac (p0 + p1 + p2) (q0 + q1 + q2 + q3) = C lam * X 0 ^ 2)
  (sA0 : ∀ i, A0.coeff i ≠ 0 → 0 ≤ i ∧ i ≤ 8)
  (sA1 : ∀ i, A1.coeff i ≠ 0 → 1 ≤ i ∧ i ≤ 8)
  (sA2 : ∀ i, A2.coeff i ≠ 0 → 1 ≤ i ∧ i ≤ 8)
  (sB0 : ∀ i, B0.coeff i ≠ 0 → 0 ≤ i ∧ i ≤ 12)
  (sB1 : ∀ i, B1.coeff i ≠ 0 → 1 ≤ i ∧ i ≤ 12)
  (sB2 : ∀ i, B2.coeff i ≠ 0 → 2 ≤ i ∧ i ≤ 12)
  (sB3 : ∀ i, B3.coeff i ≠ 0 → 2 ≤ i ∧ i ≤ 12)
  (ρ σ ε : L) (hp : ρ ≠ 0) (hσ : σ ≠ 0) (hε : ε ≠ 0)
  (t_a_1_0 : (A2.coeff 1) = ρ * ε ^ (0 : ℕ) * ((1 : L)))
  (t_a_2_2 : (A2.coeff 2) = ρ * ε ^ (1 : ℕ) * (((303) / 16 : L) + ((879) / 128 : L)
    ↪ * w + ((-389) / 128 : L) * w ^ (2 : ℕ) + ((-51) / 128 : L) * w ^ (3 : ℕ) +
    ↪ ((49) / 128 : L) * w ^ (4 : ℕ)))
  (t_a_3_4 : (A2.coeff 3) = ρ * ε ^ (2 : ℕ) * (((315) / 16 : L) + ((693) / 8 : L) *
    ↪ w + ((-147) / 8 : L) * w ^ (2 : ℕ) + ((-189) / 8 : L) * w ^ (3 : ℕ) + ((147)
    ↪ / 16 : L) * w ^ (4 : ℕ)))
  (t_a_4_6 : (A2.coeff 4) = ρ * ε ^ (3 : ℕ) * (((-79653) / 64 : L) + ((29463) / 64
    ↪ : L) * w + ((7371) / 128 : L) * w ^ (2 : ℕ) + ((-4767) / 16 : L) * w ^ (3 :
    ↪ ℕ) + ((8001) / 128 : L) * w ^ (4 : ℕ)))
  (t_a_5_8 : (A2.coeff 5) = ρ * ε ^ (4 : ℕ) * (((-33519) / 4 : L) + ((-6843) / 4 :
    ↪ L) * w + ((8181) / 8 : L) * w ^ (2 : ℕ) + ((-1713) / 2 : L) * w ^ (3 : ℕ) +
    ↪ ((-1305) / 8 : L) * w ^ (4 : ℕ)))
  (t_a_6_10 : (A2.coeff 6) = ρ * ε ^ (5 : ℕ) * (((-8805) / 8 : L) + ((-76965) / 8 :
    ↪ L) * w + ((11523) / 16 : L) * w ^ (2 : ℕ) + ((6927) / 4 : L) * w ^ (3 : ℕ) +
    ↪ ((-18579) / 16 : L) * w ^ (4 : ℕ)))
  (t_a_7_12 : (A2.coeff 7) = ρ * ε ^ (6 : ℕ) * (((389740) / 17 : L) + ((-51922) /
    ↪ 17 : L) * w + ((-48572) / 17 : L) * w ^ (2 : ℕ) + ((63239) / 17 : L) * w ^ (3
    ↪ : ℕ) + ((-6475) / 17 : L) * w ^ (4 : ℕ)))
  (t_a_8_14 : (A2.coeff 8) = ρ * ε ^ (7 : ℕ) * (((476437) / 68 : L) + ((314405) /
    ↪ 68 : L) * w + ((-174571) / 136 : L) * w ^ (2 : ℕ) + ((1574) / 17 : L) * w ^
    ↪ (3 : ℕ) + ((82783) / 136 : L) * w ^ (4 : ℕ)))
  (t_b_2_1 : (B3.coeff 2) = σ * ε ^ (0 : ℕ) * ((1 : L)))
```

```

(t_b_3_3 : (B3.coeff 3) = σ * ε ^ (1 : N) * (((101) / 8 : L) + ((293) / 64 : L) *
  ↪ w + ((-389) / 192 : L) * w ^ (2 : N) + ((-17) / 64 : L) * w ^ (3 : N) + ((49)
  ↪ / 192 : L) * w ^ (4 : N)))
(t_b_4_5 : (B3.coeff 4) = σ * ε ^ (2 : N) * (((315) / 16 : L) + ((693) / 8 : L) *
  ↪ w + ((-147) / 8 : L) * w ^ (2 : N) + ((-189) / 8 : L) * w ^ (3 : N) + ((147)
  ↪ / 16 : L) * w ^ (4 : N)))
(t_b_5_7 : (B3.coeff 5) = σ * ε ^ (3 : N) * (((-11379) / 8 : L) + ((4209) / 8 :
  ↪ L) * w + ((1053) / 16 : L) * w ^ (2 : N) + ((-681) / 2 : L) * w ^ (3 : N) +
  ↪ ((1143) / 16 : L) * w ^ (4 : N)))
(t_b_6_9 : (B3.coeff 6) = σ * ε ^ (4 : N) * (((-695299) / 48 : L) + ((-48367) /
  ↪ 48 : L) * w + ((52679) / 32 : L) * w ^ (2 : N) + ((-45661) / 24 : L) * w ^ (3
  ↪ : N) + ((-2211) / 32 : L) * w ^ (4 : N)))
(t_b_7_11 : (B3.coeff 7) = σ * ε ^ (5 : N) * (((-309895) / 8 : L) + ((-251275) /
  ↪ 8 : L) * w + ((101481) / 16 : L) * w ^ (2 : N) + ((1811) / 4 : L) * w ^ (3 :
  ↪ N) + ((-60489) / 16 : L) * w ^ (4 : N)))
(t_b_8_13 : (B3.coeff 8) = σ * ε ^ (6 : N) * (((1733348) / 13 : L) + ((-2368725)
  ↪ / 26 : L) * w + ((-184976) / 13 : L) * w ^ (2 : N) + ((1819939) / 52 : L) * w
  ↪ ^ (3 : N) + ((-569071) / 52 : L) * w ^ (4 : N)))
(t_b_9_15 : (B3.coeff 9) = σ * ε ^ (7 : N) * (((18736247) / 39 : L) + ((1087133)
  ↪ / 13 : L) * w + ((-5656777) / 78 : L) * w ^ (2 : N) + ((1993384) / 39 : L) *
  ↪ w ^ (3 : N) + ((971365) / 78 : L) * w ^ (4 : N)))
(t_b_10_17 : (B3.coeff 10) = σ * ε ^ (8 : N) * (((305098) / 17 : L) + ((5784066)
  ↪ / 17 : L) * w + ((-308335) / 17 : L) * w ^ (2 : N) + ((-950608) / 17 : L) * w
  ↪ ^ (3 : N) + ((754021) / 17 : L) * w ^ (4 : N)))
(t_b_11_19 : (B3.coeff 11) = σ * ε ^ (9 : N) * (((-149721182) / 323 : L) +
  ↪ ((26391690) / 323 : L) * w + ((21560033) / 323 : L) * w ^ (2 : N) +
  ↪ ((-24378412) / 323 : L) * w ^ (3 : N) + ((2786935) / 323 : L) * w ^ (4 : N)))
(t_b_12_21 : (B3.coeff 12) = σ * ε ^ (10 : N) * (((-91912444) / 969 : L) +
  ↪ ((-20715500) / 323 : L) * w + ((17474146) / 969 : L) * w ^ (2 : N) +
  ↪ ((-1623428) / 969 : L) * w ^ (3 : N) + ((-8783302) / 969 : L) * w ^ (4 : N)))
(hv : B0.coeff 12 ≠ 0) : False := by
-- proof body elided (see artifact)

```

9 Exact characteristic-0 descent

The proof is exact over K_5 (items (i)–(ii) below).

Lemma 9.1 (Exact obstruction). *Let $\rho \in \overline{\mathbb{Q}}$ be any root of $\mathcal{W}$, and put $K = \mathbb{Q}(\rho)$. The E_2 obstruction—no nonzero (t_1, t_2) satisfies the seven compatibility conditions—holds over K .*

Proof. The ingredients are the exact characteristic-0 computations (scripts `exact_ranks_K5.py`, `exact_obstruction_K5.py`):

- (i) *Exact characteristic-0 ranks.* The ranks are computed exactly over the quintic field $K_5 = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$ (script `exact_ranks_K5.py`): $E_4 = 17$, $E_3 = 18$, $E_2 = 12$.
- (ii) *Exact descent over K_5 .* The entire $E_4 \rightarrow E_3 \rightarrow E_2 \rightarrow 35$ -minor pipeline is redone exactly over K_5 (script `exact_obstruction_K5.py`, ~ 1 s): E_3 is solvable for every t , the left nullity of E_2 is 7, and the 35 minors span *all* binary quintics (the 35×6 coefficient matrix has rank 6 over K_5). Hence the only common zero is $t = 0$. End-to-end residuals against the original bracket equations vanish exactly. The $t = 0$ case is excluded by the vertex (12, 24) argument, which is valid in every characteristic other than 2 and 3 (it uses $2 \neq 0$ and $12 \neq 0$; see §8.3). The Lean formalization (`no_completion_K5`, listing 21, via `main_theorem_of_chart`, listing 7) quantifies over all roots of R and all torus parameters, covering all top-layer solutions by Proposition 6.1, whose $m = 7$ classification is formalized (`chartClassification_holds`, §6.4).

□

Remark 9.2 (On the derivation of $\mathcal{W}$). The polynomial $\mathcal{W}$ was originally imported from the reference data (see `W_IMPORT_DISCLOSURE.md`). It is now *independently derived*: the minimal polynomial of a_6^7 , computed from the exactly verified top-layer point over K_5 , equals the quintic V (hence $\mathcal{W}(a_6) = V(a_6^7)$) coefficient for coefficient (script `compare_V.gp`). The import caveat in `W_IMPORT_DISCLOSURE.md` is retired for $\mathcal{W}$ itself. The mod-101 reduction matches the independently computed elimination up to scalar 54 as a cross-check.

10 Conclusion

We have eliminated branches (a) and (b) of the GGHV (72, 108) candidate—case (2) of [1, Proposition 4.3]—by a computational audit:

- source-gated base objects (polygons, bracket, and φ verified against GGHV v1);
- exact layer reduction (92 equations, symbolic identity);
- top-layer determination (degree-35 eliminant, one real solution);
- descent with explicit ranks and unconditional E_3 solvability;
- minor-span obstruction $((t_1, t_2)^5)$ closing the E_2 conditions;
- exact K_5 -descent (Lean `no_completion_K5` via `main_theorem_of_chart`) covering all roots of R and all torus parameters;
- an algebraic proof of the $m = 7$ top-layer classification (Proposition 6.1 in the normalized chart), using reversed coordinates, six orbit relations and a Sylvester resultant (§6.4). It is formalized in Lean, so the NewtonNF2 form of Theorem 1.1 is the unconditional kernel-checked theorem `main_theorem`.

11 Prior work, attribution, and independence

Prior work. F. Santibañez-Leal’s CAOS records EXP-061 and EXP-065 treat exactly these cases with sampled certificates, later downgraded by his own audit. We cite them here for completeness.

Attribution. The layer reduction strategy ($u = xy^2$, five triangular layers), the identification of the Davenport–Stothers top layer, the descent architecture, and the modular certificate approach originate in the initial computational package (`branch_ab_elimination.zip`, MD5 `0cfa49d0...`). Its full-formulation certificate at $p = 10^9 + 9$ and $p = 10^9 + 21$, with lifting argument, is a second, independent route to the same obstruction (see the Appendix). This work reimplements the reduction independently in SINGULAR (C1) and recomputes the descent, but the overall strategy is not original to us.

Characteristic-0 core. The characteristic-zero computations—the Belyi counts (`belyi_counts_m357.py`), the exact top-layer point over K_5 (`e5_exact_K5.json`), the derivation of $\mathcal{W}$ (`compare_V.gp`), the exact ranks (`exact_ranks_K5.py`), the exact $E_4 \rightarrow E_3 \rightarrow E_2$ descent (`exact_obstruction_K5.py`), and the msolve-parametrization check (`verify_msolve_param.py`)—are documented in the scripts (see their attribution headers). Tools: python-flint 0.9.0 (exact K_5 arithmetic), sympy 1.14.0, PARI/GP 2.15.4, Singular 4.3.2. The independence case rests on the artifacts, not on who ran them: the Lean kernel checks every formalized proof; the Singular Gröbner computations (`a816_full.sing`, `chart_system_Q.sing`) can be rerun from the archive; and the Belyi counts were verified by an independent Murnaghan–Nakayama implementation.

Independent blind replication (completed 2026-09-25). An independent blind replication was

completed by Claude (Anthropic, via Claude Code in a cloud container). The agent received only a short brief—the degree pair $(72, 108)$, the instruction to work from GGHV Proposition 4.3, and no access to this work or its scripts—plus GGHV v1 itself. Verdict: **DO NOT EXIST**. The proof uses weight grading, a Belyi-map count (at most 35 complex points via Riemann’s existence theorem and the Frobenius character formula), exact modular certificates at $p = 32003$ and $p = 536870909$ (Singular, Macaulay2, msolve), and Hensel lifting. It proves the stronger lower-edge rigidity: $a_{8,16} = b_{12,24} = 0$. Materials: [tier1_blind_gghv43_nf2/](#) [8], committed as 37c73b67 on 2026-09-25.

AI usage. Two AI systems contributed to this work. Claude (Anthropic) served as pre-submission reviewer and computed the characteristic-zero verification scripts (`belyi_counts_m357.py`, `exact_ranks_K5.py`, `exact_obstruction_K5.py`, `verify_msolve_param.py`; see their attribution headers), the chart audit (`audit_chart.py`), the $a_{8,16} = 0$ Gröbner computation (`a816_full.sing`), the Murnaghan–Nakayama implementation (`frobenius.py`), and the independent Lean build. During the v17 review Claude also found the algebraic proof of the $m = 7$ top-layer classification in §6.4 (the reversed coordinates, the six orbit relations and their certificates) and wrote its Lean formalization (`lean/Jacobian/ChartProof`, with the generator `lean/certgen/chartproof`). Claude further performed the independent blind replication described above (see [tier1_blind_gghv43_nf2/](#) commit history). Muse (Meta), the author’s personal AI assistant, generated the Lean listings mechanically from committed source (`gen_listings.py`) and assisted with manuscript drafting; all such outputs were verified by execution or by the Lean kernel. The author (B. D. McCrary) directed the investigation, verified the results, and takes responsibility for the manuscript.

Remark 11.1 (What remains). The application of Theorem 1.1 to the Jacobian conjecture, Corollary 1.2, is conditional on [1, Proposition 4.3] (Remark 1.3). The remaining open case is branch (c) (case (1) of Proposition 2.1), whose Newton polygons share the same top-layer E_5 equation and are amenable to the same machinery with additional lower layers. The status of the $(72, 108)$ candidate as a whole remains underdetermined pending branch (c).

References

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- [7] B. D. McCrary, Muse, *The $(72, 108)$ candidate for the planar Jacobian conjecture: a containment note*, Zenodo [doi:10.5281/zenodo.22941057](https://doi.org/10.5281/zenodo.22941057) (2026).

[8] Claude (Anthropic), *Independent blind replication: GGHV Proposition 4.3, normal form (2)*, [tier1_blind_gghv43_nf2/](#) in B. D. McCrary, *2D-Jacobian-Conjecture-Workspace*, GitHub (2026), commit 37c73b67 (2026-09-25).

A Computational archive

All scripts used in this audit are archived in `tier2_branch_ab_scripts.zip`:

- `C1_layer_reduction.sing` — exact 92-equation layer reduction (symbolic identity);
- `source_gate_verify.py` — asserts GGHV v1 lines 33, 492–495, 596–600;
- E_4 (an 18×19 matrix of rank 17), kernel 2 — verified within `C5_E2_obstruction_corrected.py` (with $A_2 = u\alpha$, $B_3 = u^2\beta$);
- `C5_E2_obstruction_corrected.py` — E_3 (a 19×20 matrix of rank 18) and $v^\top K \equiv 0$ (with residual asserts);
- Top-layer elimination, SINGULAR factorization over $\mathbb{Q}$, discriminant and Sturm computations;
- E_2 rank verification (injectivity of the 19×12 operator).

The generator of the Lean proof of Theorem 6.5 (§6.4) ships with the Lean project, in `lean/certgen/chartproof/`. It finds the orbit relations by exact interpolation, computes the certificates by weighted-homogeneous linear algebra over $\mathbb{Q}$, and verifies all 105 step identities in SymPy; `regenerate_chartproof.sh` checks byte-identical regeneration. The E_2 seven-condition expansion ($b_i + s_1 M_i + s_2 L_i = 0$) and the 35-minor Gröbner basis over $\mathbb{F}_{101}$ (§8) are computed in the saved, rerunnable script `C5_E2_obstruction_corrected.py`; the log `C5_E2_obstruction_corrected.log` is archived. Modular computations use $p = 101$; characteristic-zero claims (irreducibility over $\mathbb{Q}$, discriminant nonzero, exactly one real root) are over $\mathbb{Q}$. The earlier E_4/E_3 scripts using unscaled (α, β) are superseded and excluded from the archive.

B Original certificate (second route)

Attribution. The following certificate was computed by Claude (Anthropic) during review, as part of the original package `branch_ab_elimination.zip`. It is included here as a second, independent route to the same obstruction.

Method. Full-formulation (remainder-based) certificate: constructs the E_2 conditions via polynomial remainder rather than left-null-space, uses FLINT for linear algebra (rather than hand-coded Gaussian elimination), and runs at $p = 10^9 + 9$ and $p = 10^9 + 21$ (rather than $p = 101$).

Shared inputs. Both routes share: GGHV’s normal form (Proposition 4.3), the exact layer reduction formula ($u = xy^2$, five triangular layers), and the eliminating polynomial $\mathcal{W}$.

Result. At $p = 1000000009$: 7 simple roots of $\mathcal{W}$ in $\mathbb{F}_p$, 2 certified with `E4_rank=17`, `E3_rank=18`, `E3_consistent_at_nodes=True`. At $p = 1000000021$: 2 simple roots, 2 certified. All show `certified=True`.

Coverage of all top-layer solutions. The Lean theorems `no_completion_K5` (listing 21) and `main_theorem_of_chart` (listing 7) quantify over all roots w of R and all torus parameters $\rho, \sigma, \varepsilon \neq 0$, so “no top-layer solution admits a completion” holds for every solution of the top-layer system by Proposition 6.1 (for $m = 7$ formalized as `chartClassification_holds`),

not just the K_5 -point. The exact K_5 computation (scripts `exact_ranks_K5.py`, `exact_obstruction_K5.py`) is an independent cross-check of the same statement.

Scripts: `cert_branch_ab.py`, `cert_setup.py`; log: `cert_branch_ab.log`. All archived in `tier2_branch_ab_scripts.zip`.